

Supplementary Figures

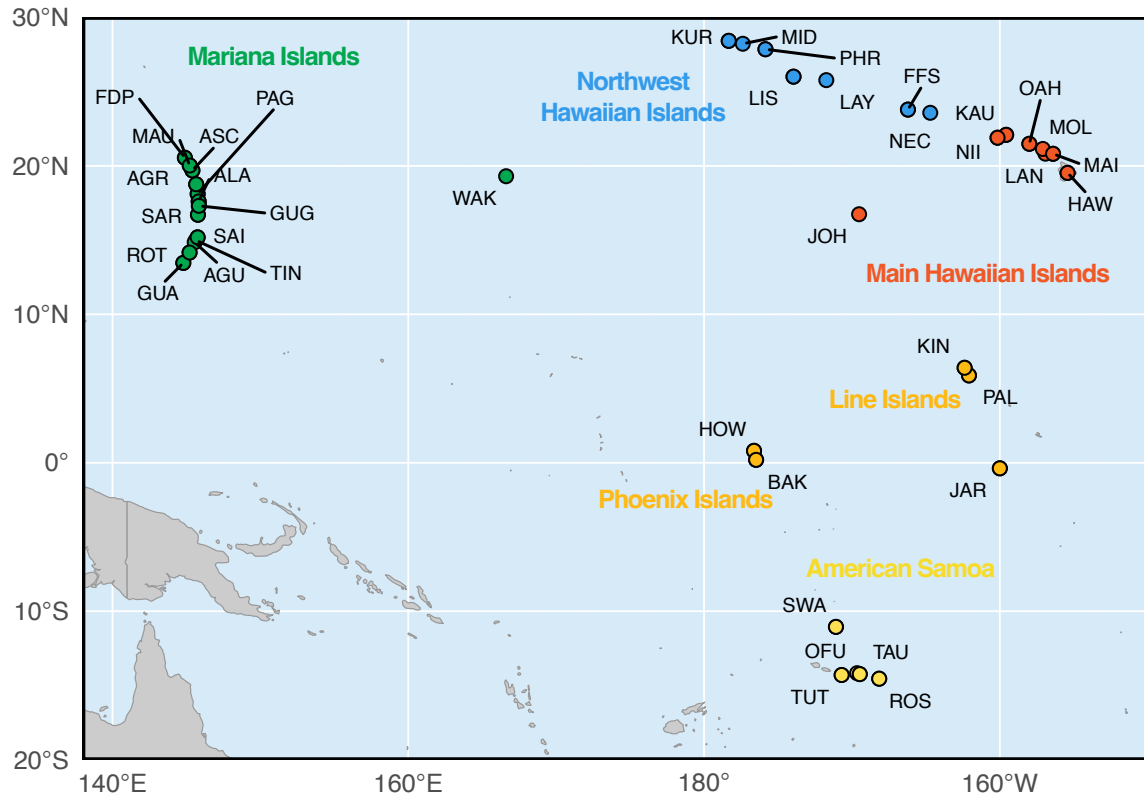


Fig. S1 | Map of islands analysed for IME and fish energy flux across the tropical Pacific. IME strength was measured at 34 island or island complexes, which were Saipan (Saipan, Tinian, Aguijan), Maui (Maui, Molokai, Lanai), and Tau (Tau, Ofu & Olosega). Fish surveys were conducted at 34 islands, separately each each complex island. We excluded Johnston, Laysan, and Wake due to lack of site-level bathymetry.

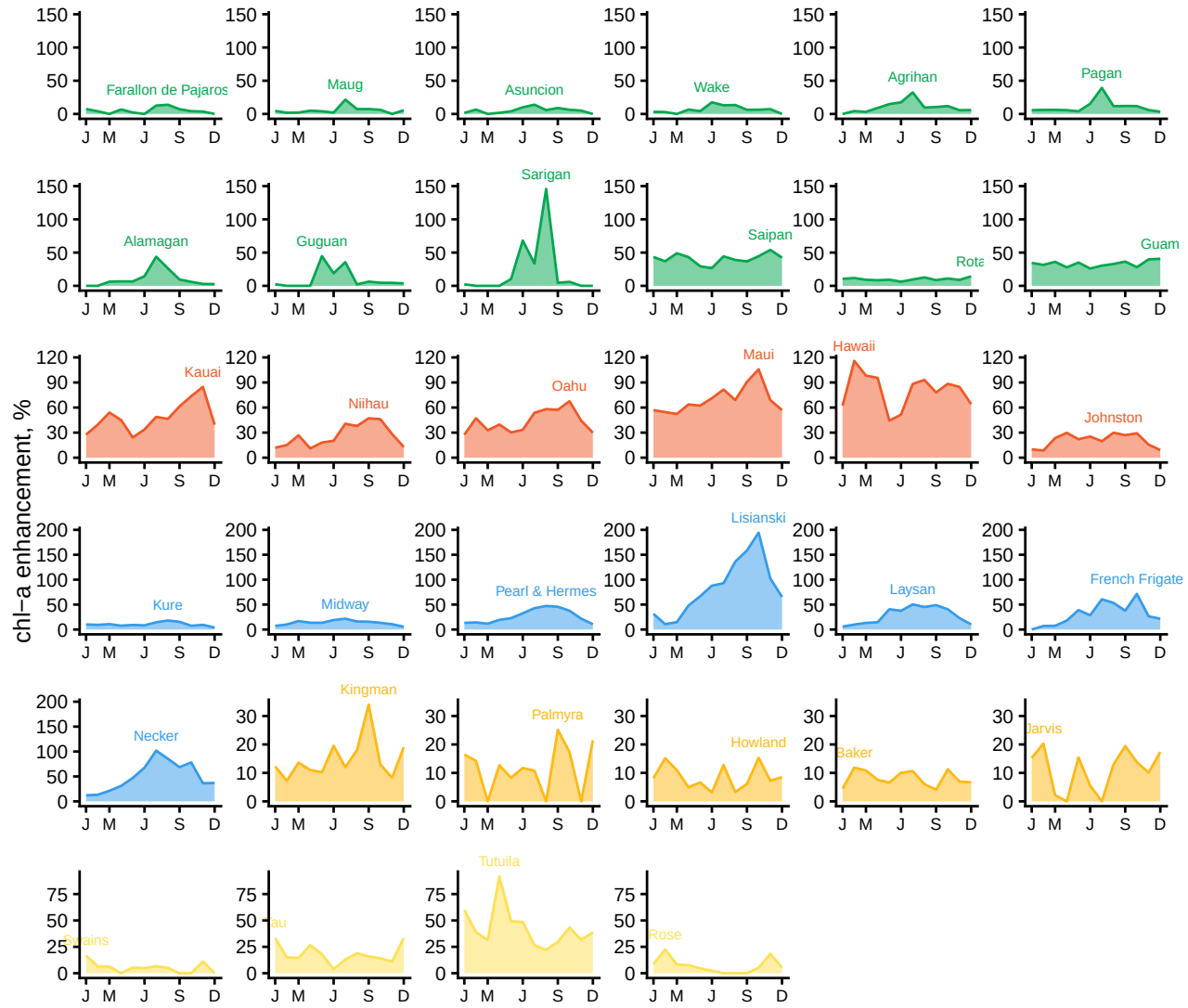


Fig. S2 | Seasonal variation in IME strength at each island. Lines are mean-monthly chl-a enhancement from January to December, ordered and coloured by island group.

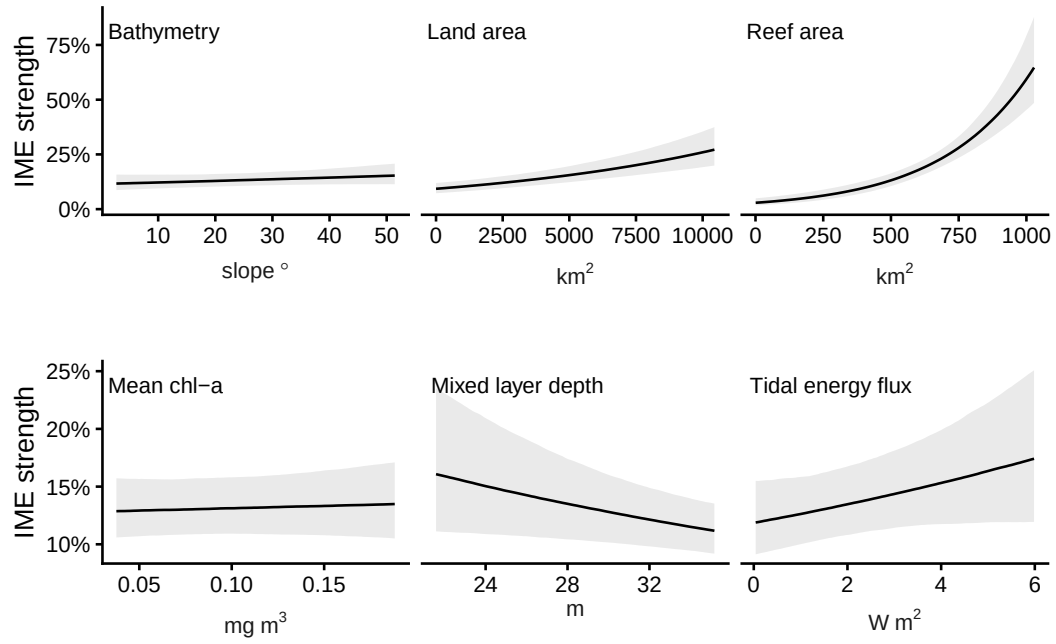


Fig. S3 | Spatial effects on chl-a enhancement for geomorphic features (top) and average oceanographic conditions (bottom) Lines are median posterior predictions ($\pm$ 95% certainty intervals). Note different y-axis values between static and dynamic covariates.

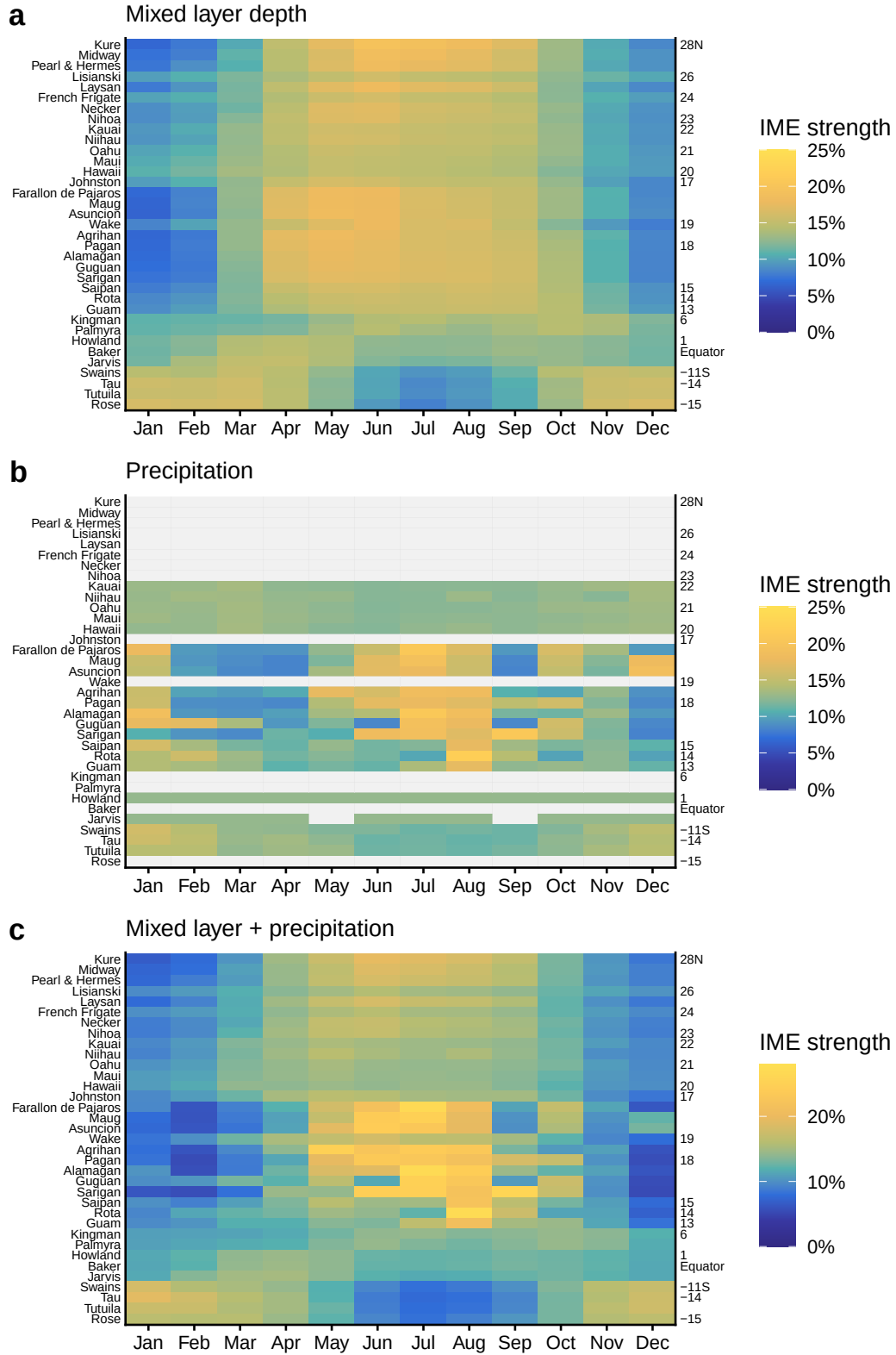


Fig. S4 | Heatmap of seasonal effects on chl-a enhancement for a mixed layer depth, b precipitation and c both covariates combined. Islands with less than 1 mm monthly rainfall are shown in grey. All values are median posterior predictions excluding effects of other covariates.

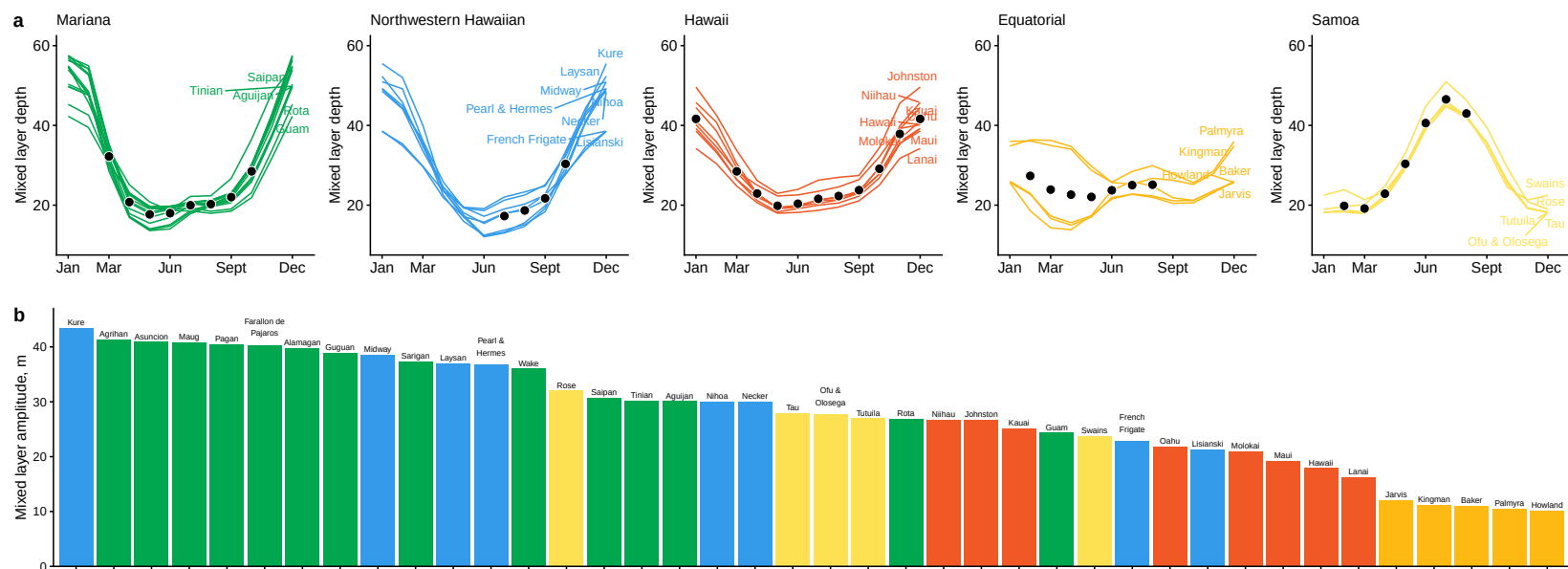


Fig. S5 | Seasonal variability in the mixed layer depth at each island. In a, lines are predicted mixed layer depth at each island from a General Additive Model with smoothers for time and month (cyclic), overlaid with points indicating timing of coral reef fish surveys.

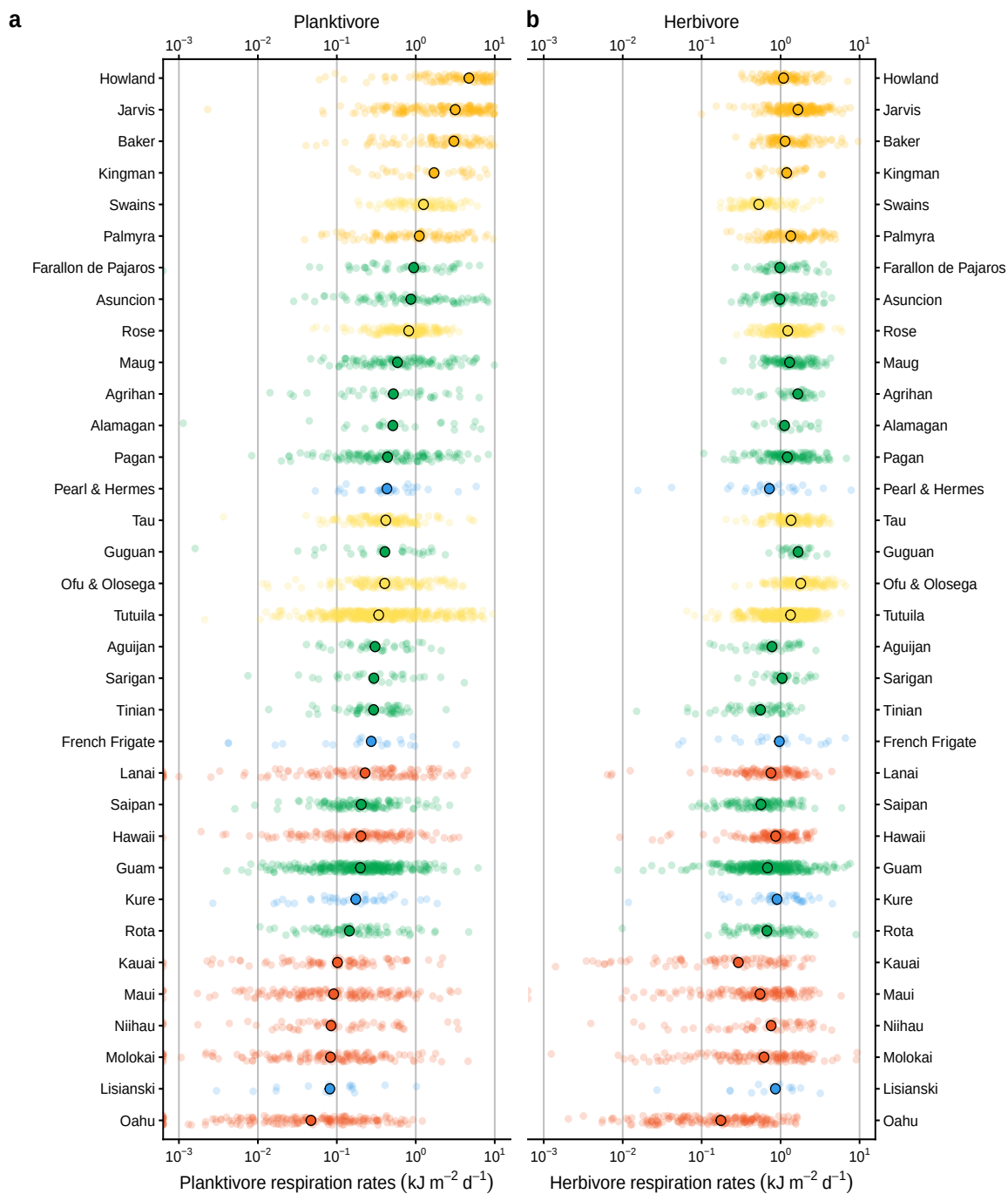


Fig. S6 | Respiration rate of reef-associated planktivores on coral reefs at 34 Pacific islands and atolls. Each point is the median respiration at each island, with jitter showing unique sites ($n = 3,622$).

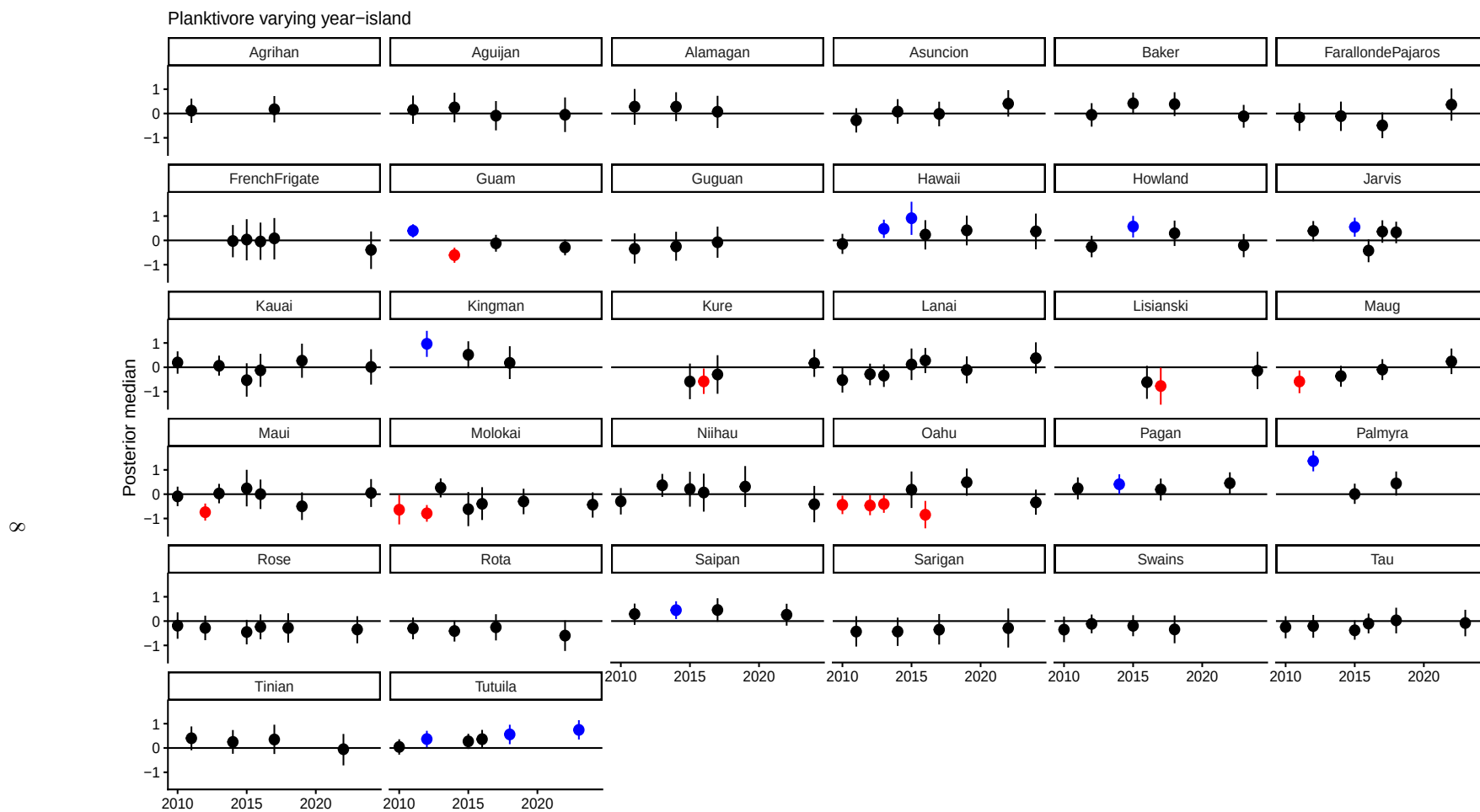


Fig. S7 | Effect of survey year on planktivore respiration rate. Points are posterior medians with 95% certainty intervals, coloured when the posterior did not overlap zero (red = negative, blue = positive).

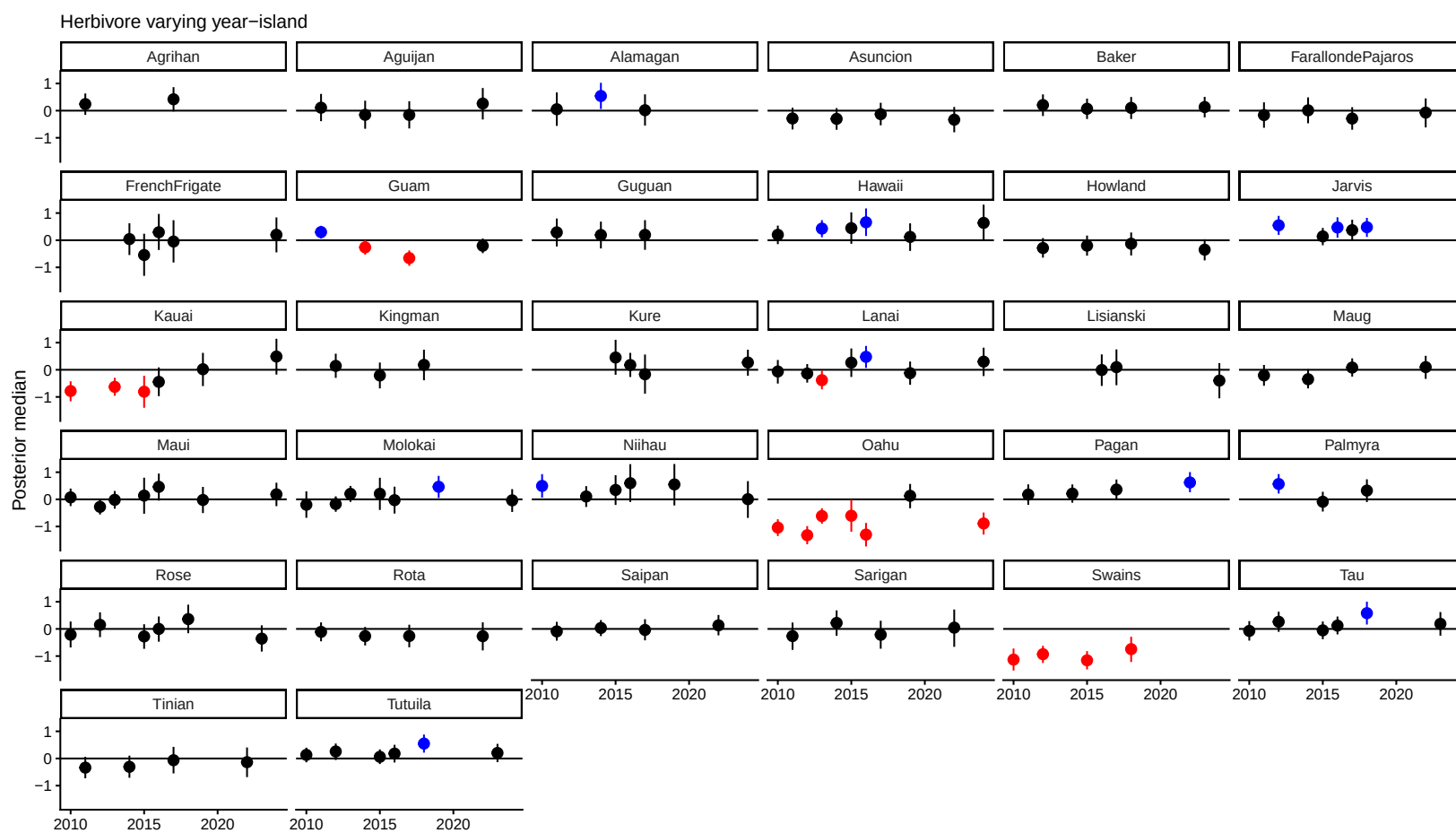


Fig. S8 | Effect of survey year on herbivore respiration rate. Points are posterior medians with 95% certainty intervals, coloured when the posterior did not overlap zero (red = negative, blue = positive).

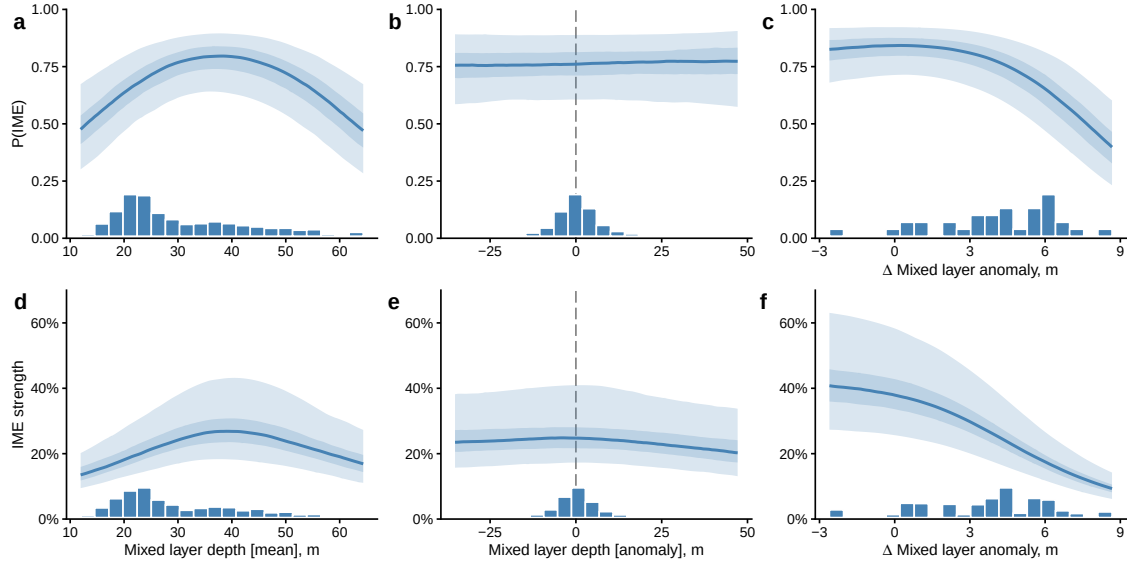


Fig. S10 | Effects of mixed layer depth covariates on IME dynamics from 1997-2025. Top-row is probability of detecting an IME and bottom-row is IME strength (%), for mean mixed layer depth (a, d), mixed layer depth anomaly (b, e), and the mixed layer change from 1993-2026 (c, f). Lines are median posterior estimates with 95% and 50% certainty intervals. Histograms show underlying data distribution of each covariate along x-axis.

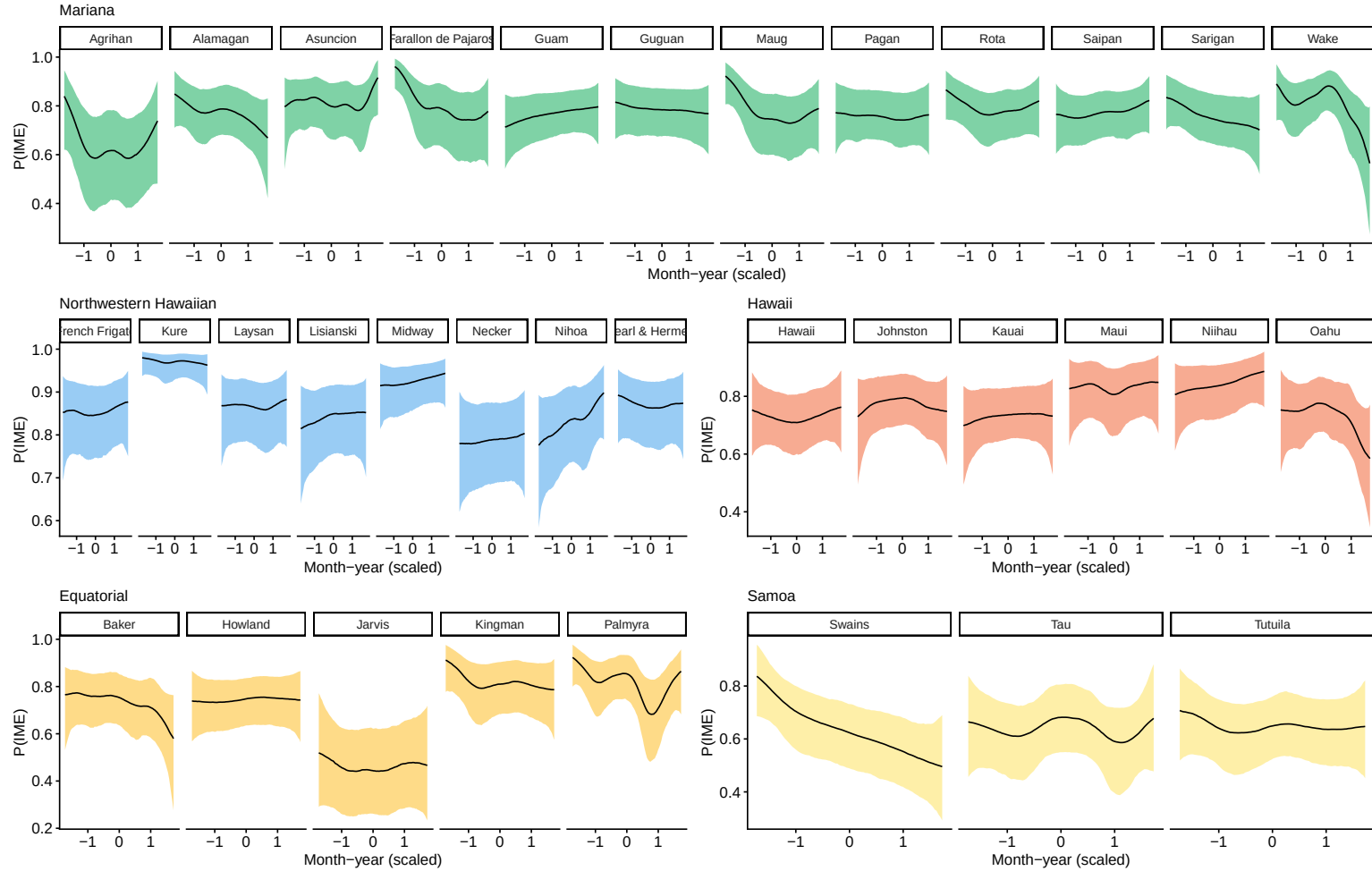


Fig. S11 | Temporal change in the probability of IME occurrence at each island from 1997-2025. Lines are median posterior estimates with 95% and 50% certainty intervals, excluding effects of mixed layer covariates and month smoother.

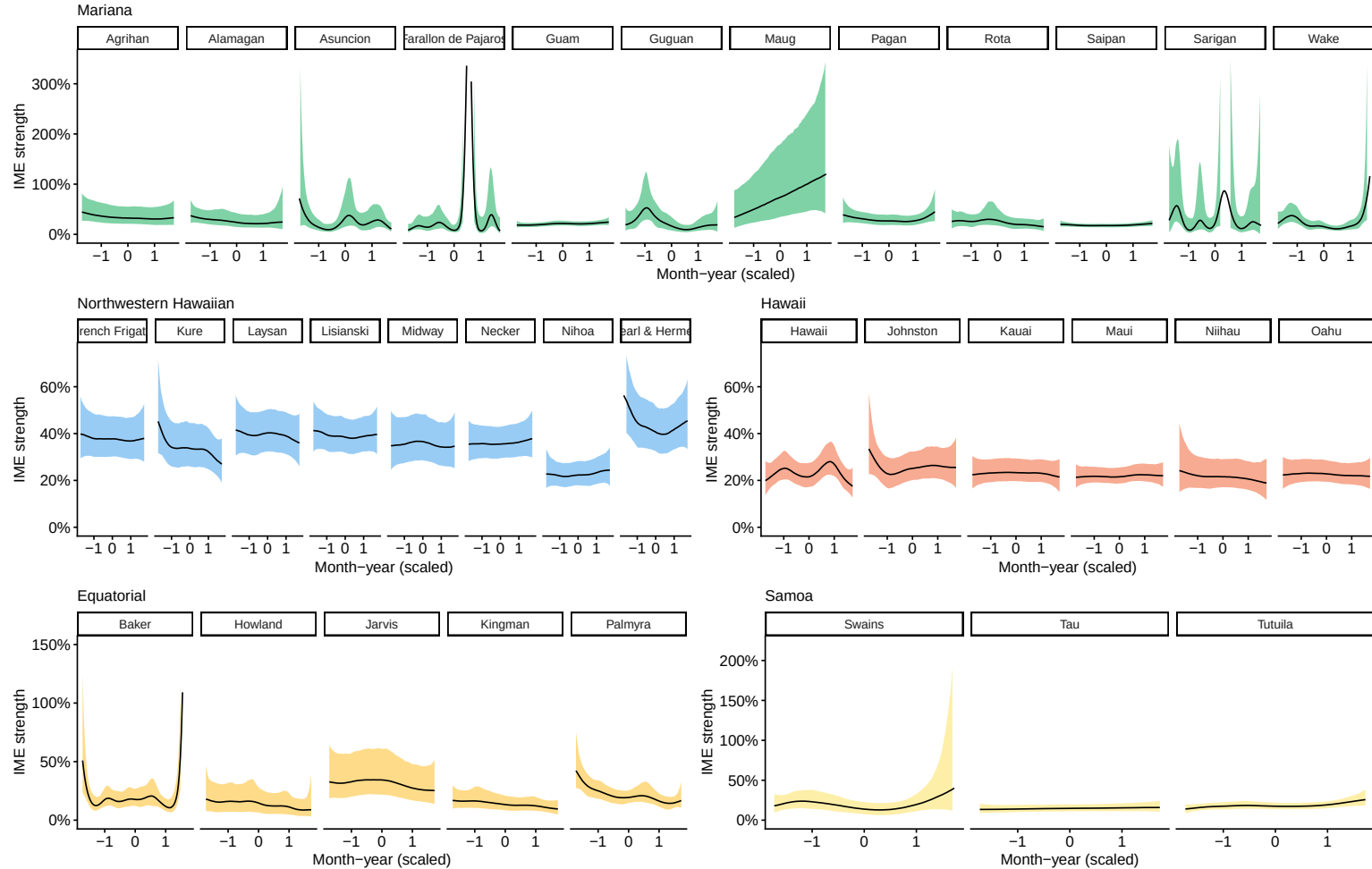


Fig. S12 | Temporal change in the IME strength at each island from 1997-2025. Lines are median posterior estimates with 95% and 50% certainty intervals, excluding effects of mixed layer covariates and month smoother.

	Covariates	Δelpd	$\text{SE}(\Delta\text{elpd})$	elpd	$\text{SE}(\text{elpd})$	p_{loo}	$\text{se}_{p_{\text{loo}}}$	looic	se_{looic}
	MLD_vars + time + month	0.00	0.00	-7101.91	49.58	276.47	6.64	14203.83	99.16
IME detect	MLD_vars	-227.59	36.22	-7329.51	33.11	270.96	3.14	14659.01	66.21
	time + month	-361.01	39.64	-7462.92	27.69	206.48	2.04	14925.84	55.38
	MLD_vars + time + month	0.00	0.00	5713.96	79.43	450.04	20.96	-11427.92	158.87
IME strength	MLD_vars	-491.53	41.01	5222.43	74.68	486.53	17.23	-10444.86	149.35
	time + month	-793.65	46.33	4920.30	79.15	460.57	20.12	-9840.61	158.31

Table S1 | Model comparison using leave-out-out cross-validation (LOO). Models fitted to probability of IME detection (IME detect; Bernoulli distributed) and IME strength (chl-a enhancement, %; Gamma-distributed), for nested models with smoothers for the mixed layer climatology (monthly mean), anomaly, and long-term trend of time and month (MLD_vars), and temporal smooths (time + month). Top-ranked models have the lowest Bayesian LOO of expected log pointwise predictive density (elpd). Additional estimates provided for effective number of parameters (p_{loo}) and LOO information criterion (looic).